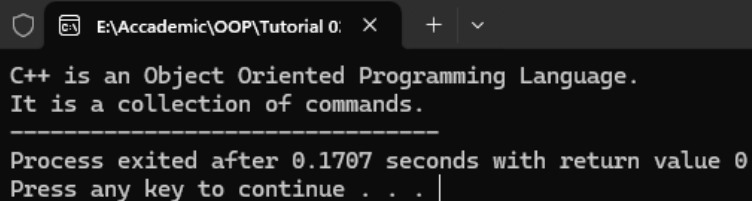


Tutorial 02

01)

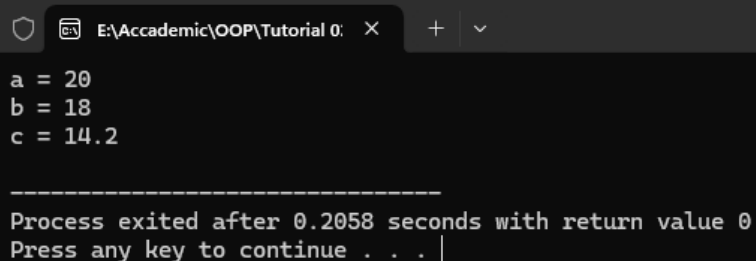
```
1 #include <iostream>
2
3 using namespace std;
4
5 int main(){
6
7     cout<< "C++ is an Object Oriented Programming Language. \n";
8     cout<< "It is a collection of commands.";
9     return 0;
10 }
```



```
E:\Accademic\OOP\Tutorial 0: X + v
C++ is an Object Oriented Programming Language.
It is a collection of commands.
-----
Process exited after 0.1707 seconds with return value 0
Press any key to continue . . . |
```

02)

```
1 #include <iostream>
2
3 using namespace std;
4
5 int main(){
6
7     int a = 20, b = 18;
8     float c = 14.2;
9
10    cout<< "a = " << a << "\n";
11    cout<< "b = " << b << "\n";
12    cout<< "c = " << c << "\n";
13    return 0;
14 }
```



```
E:\Accademic\OOP\Tutorial 0: X + v
a = 20
b = 18
c = 14.2
-----
Process exited after 0.2058 seconds with return value 0
Press any key to continue . . . |
```

03)

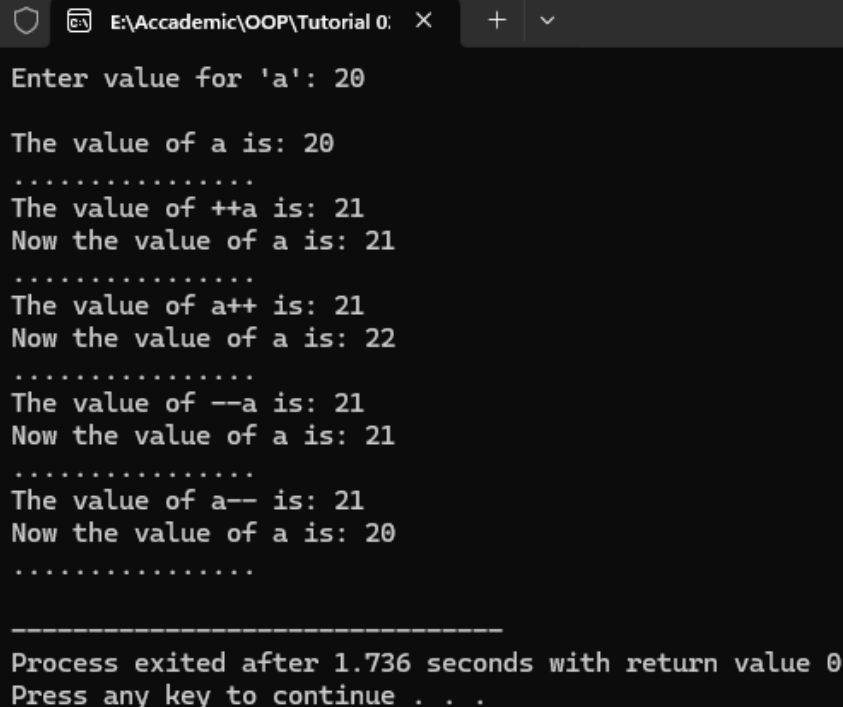
```
1  #include <iostream>
2
3  using namespace std;
4
5  int main(){
6
7      int arr[4];
8      int i,j;
9
10     for(i=0; i<=4; i++){
11         cout << "Enter Your number "<<i+1<<" : ";
12         cin >> arr[i];
13     }
14
15     cout<< "Forward: ";
16     for (i=0; i<=4; i++){
17         cout<< arr[i] << " ";
18     }
19
20     cout<< "\nReverse: ";
21     for (j=4; j>=0; j--){
22         cout<< arr[j] << " ";
23     }
24     return 0;
25 }
```

```
Enter Your number 1: 1
Enter Your number 2: 2
Enter Your number 3: 3
Enter Your number 4: 4
Enter Your number 5: 5
Forward: 1 2 3 4 5
Reverse: 5 4 3 2 1
```

```
-----
Process exited after 4.702 seconds with return value 0
Press any key to continue . . . |
```

04)

```
1  #include <iostream>
2
3  using namespace std;
4
5  int main(){
6
7      int a;
8      cout << "Enter value for 'a': ";
9      cin >> a;
10
11     cout << "\nThe value of a is: " << a << "\n" << "....." << "\n";
12
13     cout << "The value of ++a is: " << ++a << "\n";
14     cout << "Now the value of a is: " << a << "\n" << "....." << "\n";
15
16     cout << "The value of a++ is: " << a++ << "\n";
17     cout << "Now the value of a is: " << a << "\n" << "....." << "\n";
18
19     cout << "The value of --a is: " << --a << "\n";
20     cout << "Now the value of a is: " << a << "\n" << "....." << "\n";
21
22     cout << "The value of a-- is: " << a-- << "\n";
23     cout << "Now the value of a is: " << a << "\n" << "....." << "\n";
24
25     return 0;
26 }
```



```
Enter value for 'a': 20

The value of a is: 20
.....
The value of ++a is: 21
Now the value of a is: 21
.....
The value of a++ is: 21
Now the value of a is: 22
.....
The value of --a is: 21
Now the value of a is: 21
.....
The value of a-- is: 21
Now the value of a is: 20
.....

-----
Process exited after 1.736 seconds with return value 0
Press any key to continue . . .
```

05)

```
1  #include <iostream>
2  using namespace std;
3
4  int main() {
5      int arr[3];
6      int i;
7
8      for (i = 0; i < 3; i++) {
9          cout << "Enter value " << i + 1 << ": ";
10         cin >> arr[i];
11     }
12
13     int max = arr[0];
14     for (i = 1; i < 3; i++) {
15         if (arr[i] > max) {
16             max = arr[i];
17         }
18     }
19
20     cout << "Maximum value is: " << max << endl;
21
22     return 0;
23 }
24
```

```
Enter value 1: 2
Enter value 2: 6
Enter value 3: 75
Maximum value is: 75
```

```
-----
Process exited after 4.144 seconds with return value 0
Press any key to continue . . . |
```

06)

```
1  #include <iostream>
2
3  using namespace std;
4
5  int main(){
6
7      for (int i=1; i<=7; i++){
8          for(int j=1; j<=i; j++){
9              cout << j;
10         }
11
12         for(int k=1; k<=7-i; k++){
13             cout << "*";
14         }
15         cout << "\n";
16     }
17     return 0;
18 }
```

```
1*****
12*****
123*****
1234***
12345**
123456*
1234567
```

```
-----
Process exited after 0.1951 seconds with return value 0
Press any key to continue . . . |
```

08)

```
1  #include <iostream>
2
3  using namespace std;
4
5  int main(){
6
7      for (int i=30; i>=10; i--) {
8          cout << "Square of " << i << " = " << i*i << "\n";
9      }
10
11     return 0;
12
13 }
```

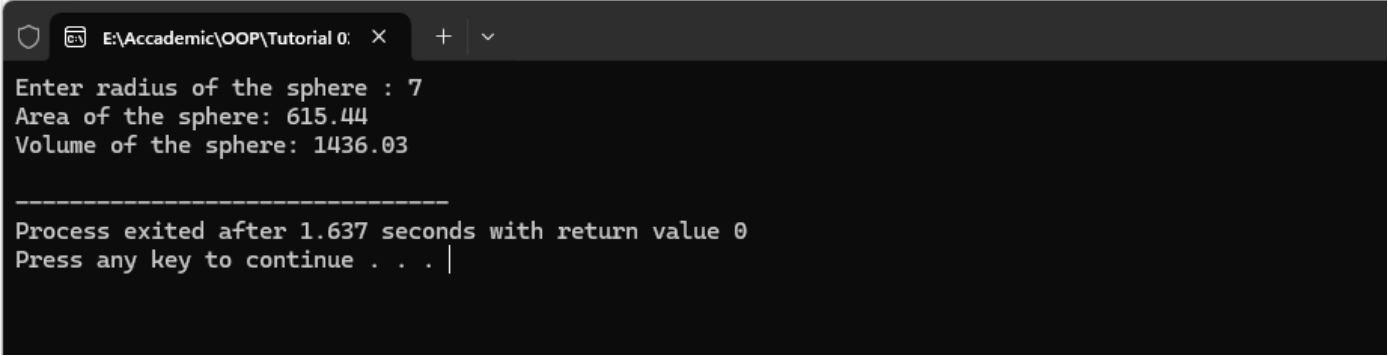
```

Square of 30 = 900
Square of 29 = 841
Square of 28 = 784
Square of 27 = 729
Square of 26 = 676
Square of 25 = 625
Square of 24 = 576
Square of 23 = 529
Square of 22 = 484
Square of 21 = 441
Square of 20 = 400
Square of 19 = 361
Square of 18 = 324
Square of 17 = 289
Square of 16 = 256
Square of 15 = 225
Square of 14 = 196
Square of 13 = 169
Square of 12 = 144
Square of 11 = 121
Square of 10 = 100
```

```
-----
Process exited after 0.2011 seconds with return value 0
Press any key to continue . . .
```

09)

```
1  #include <iostream>
2
3  using namespace std;
4
5  int main(){
6
7      const float pi = 3.14;
8      float radius, area, volume;
9
10     cout << "Enter radius of the sphere : ";
11     cin >> radius;
12
13     cout << "Area of the sphere: " << 4*pi*radius*radius<<"\n";
14     cout << "Volume of the sphere: " << (4.0/3)*pi*radius*radius*radius<<"\n";
15
16     return 0;
17 }
```



The screenshot shows a terminal window with a dark background. The title bar at the top indicates the file path 'E:\Accademic\OOP\Tutorial 0:'. The program's output is displayed in white text. It prompts the user to enter the radius of a sphere, which is entered as 7. It then calculates and displays the area of the sphere as 615.44 and the volume as 1436.03. After the calculations, a horizontal line is drawn, and the program reports that it exited after 1.637 seconds with a return value of 0. It ends with a prompt for the user to press any key to continue.

```
E:\Accademic\OOP\Tutorial 0:  ×  +  ▾
Enter radius of the sphere : 7
Area of the sphere: 615.44
Volume of the sphere: 1436.03

-----
Process exited after 1.637 seconds with return value 0
Press any key to continue . . . |
```

10)

```
1  #include <iostream>
2  using namespace std;
3
4  void reverseArray(int arr[], int size) {
5      int start = 0;
6      int end = size - 1;
7
8      while (start < end) {
9          int temp = arr[start];
10         arr[start] = arr[end];
11         arr[end] = temp;
12
13         start++;
14         end--;
15     }
16 }
17
18 int main() {
19     int Array[7] = {4, 2, 7, 5, 8, 1, 6};
20
21     cout << "Original Array: ";
22     for (int i = 0; i < 7; i++) {
23         cout << Array[i] << " ";
24     }
25     cout << endl;
26
27     reverseArray(Array, 7);
28
29     cout << "Reversed Array: ";
30     for (int i = 0; i < 7; i++) {
31         cout << Array[i] << " ";
32     }
33     cout << endl;
34
35     return 0;
36 }
37
```

```
E:\Accademic\OOP\Tutorial 0: X + -
Original Array: 4 2 7 5 8 1 6
Reversed Array: 6 1 8 5 7 2 4

-----
Process exited after 0.2049 seconds with return value 0
Press any key to continue . . .
```